The saturated dual resolvent

A specialist dossier on the critical bordered block $R^\dagger$

Stefan Hamann

August 29, 2026

Abstract

On a family of explicit finite discrete measures, a particle–hole duality identifies a two-measure subordination inequality with the spectral condition $R \succ \frac{1}{2}I$ on a dual hole kernel. A rank-one border of that kernel produces a single object $R^\dagger$ whose positivity $R^\dagger \succ \frac{1}{2}I$ is equivalent to the original inequality *and* a terminal budget condition, and whose critical three-by-three Schur block carries both. A constrained equilibrium problem on the same dual weight places the binding pair of nodes astride a saturated hard-edge block. All identities below are finite linear algebra; all numerical values are a frozen census on a finite family. No asymptotic theorem is proved. After the two-rank inertia cut and the matrix-Weyl sequence (rounds 367–373) the one open question is the *minimal inertia form* Problem 1: an integer mode count and the sign of an $O(1)$ 2×2 source, not a margin asymptotic. The older saturation/RHP question is retained as the stronger variant Problem 2. This document is a problem statement for specialists, not a claim, and nothing in it is evidence for or against the Riemann hypothesis.

Status and grades. Every assertion is marked EXACT (finite identity, including rational arithmetic), THEOREM-GRADE (finite-matrix theorem on measured inputs), LEAN-VERIFIED (sorry-free in `rh/lean/`, kernel-checked), CENSUS (measured on a frozen finite family; no extrapolation), HYPOTHESIS (unproved analytic statement), or REFUTED (named internal route closed by a sealed probe). The construction of the underlying measures (μ, ν) is given in full in the companion note [4]; the present dossier does not repeat it. Uvarov-transform language, the discrete-hard-edge placement, and the Lubinsky-class exclusion are recorded there (§frozen-placement, §frozen-dualhole) and are cited, not duplicated. Machine records: sealed probes of rounds 356, 359, 360, 362, 363, 366, 367 and 369–372 (repository `experiments/tfpt-discovery/`); Lean transcription round 373 (`RH/Haynsworth.lean`, `RH/DualResolvent.lean`); graph-resolvent consolidation round 412 (`RH/GraphResolvent.lean`); edge-balance consolidation round 426 (`RH/EdgeBalance.lean`).

1 Setup

Fix a prime power z (the window anchor). A triangular ("tent") test function of half-width Δ, determined by z and a fixed oversampling parameter, samples the von Mangoldt comb and an archimedean kernel, producing an even trigonometric polynomial f — the symbol of a circulant. Folding the symbol onto $[-1, 1]$ by $x_j = \cos\theta_j$ yields a signed discrete measure on the cosine grid, split by sign into two positive measures μ and ν with disjoint supports in $[-1, 1)$ and S atoms in total [4, construction]. Write $N = \lceil S/2 \rceil$ and $Y = \operatorname{supp}\nu$. Across the whole family the grid is unculled, so $S = 2N - 1$ [exact]: half-filling is Borodin's complementary rank condition, not an extra hypothesis.

The running instance is $z = 16$: $S = 367$, $N = 184$, $|Y| = 104$. The binding pair of ν-atoms sits at grid (fold) indices $(2, 4)$, the two rightmost odd nodes of the cosine lattice. The inequality under study, denoted L^*, asks

$$\int p^2\,\mathrm{d}\nu \;<\; \int p^2\,\mathrm{d}\mu \qquad \text{for every real polynomial } p \neq 0 \text{ with } \deg p < N.$$

Equivalently, the ν-sampling matrix E_N in the μ-orthonormal frame satisfies $\lambda_{\max}(\mathsf{E}_N) < 1$. An augmented form $L^\dagger$ adjoins a rank-one border (a budget-normalized smooth test functional) and asks $\lambda_{\max}(\mathsf{E}_N + bb^T/B) < 1$. Both statements are open for the unbounded family. On every tested window the inequalities hold, with spectral margins decaying as a power of N; structurally destroyed comparison inputs fail already at a small fraction of the critical degree [4]. No statement here claims either inequality.

2 Definition of $R^\dagger$

Lift (μ,ν) to the positive measure $\eta = \mu+\nu$ with weights $u_j = |\widetilde{w}_j|$ on the union $X = \operatorname{supp}\eta$. Let P_X be the monic node polynomial of X. Borodin's reciprocal weight [1] is $u_j^\vee = 1/(u_j\,P_X'(x_j)^2)$. On this cosine grid the dual weight has the closed form

$$u_j^\vee \;\propto\; \frac{c_j\,(1-x_j)}{|f(\theta_j)|}\,, \tag{1}$$

where c_j is the folding weight ($c_j = \frac{1}{2}$ at the left endpoint, else 1): the original weight carries $+\log|f|$, the dual weight $-\log|f|$ [exact, three evaluation routes]. Let $\Pi_{N-1}^{u^\vee}$ be the rank-$(N-1)$ Christoffel–Darboux projection kernel of $u^\vee$ on X, dressed by $\sqrt{u^\vee}$, and let R be its restriction to Y.

Let b be the border functional in the μ-orthonormal frame, $B > 0$ the budget, $\beta = b/\sqrt{B}$, and v the node-side image of the border under the μ-Christoffel–Darboux kernel ($v_j = \sqrt{\nu_j/B}\int K_\mu(y_j,t)\,\mathrm{d}\sigma(t)$). Write $\gamma = \beta^T\beta$ and $\widetilde{v} = Dv$, with D the diagonal sign matrix of P_X'.

Definition 1 (Bordered dual resolvent)**.** *The* augmented dual kernel *is the* $(|Y|+1)\times(|Y|+1)$ *matrix*

$$R^\dagger \;=\; \begin{pmatrix} R^{-1} & \widetilde{v} \\ \widetilde{v}^T & 1+\gamma \end{pmatrix}^{-1}. \tag{2}$$

The extra coordinate is the border; the leading principal block is a rank-one Sherman–Morrison deformation of R (Proposition 2 (A4) below).

Route (i) — adjoining a virtual atom and taking the $(S+1)$-point Borodin kernel — is available on a single-atom toy border (half-filling shifts $S = 2N-1 \mapsto S^\dagger = 2N$, dual rank $N-1 \mapsto N$) and agrees with (2) up to sign conjugation [exact, rational arithmetic]. On the real windows the border is a signed smooth-comb measure with S atoms, every one of which collides with a grid node, and is not a point evaluation (representability spread 56.8 on the running instance). The bordered-resolvent form is therefore the general object.

3 Exact duality formulae

Write margin $= 1 - \lambda_{\max}(\mathsf{E}_N)$ and margin$^\dagger = 1 - \lambda_{\max}(\mathsf{E}_N + bb^T/B)$.

Proposition 1 (Particle–hole dictionary [1])**.** *At half-filling $S = 2N-1$,*

$$L^* \iff \lambda_{\max}(\mathsf{E}_N) < 1 \iff R \succ \tfrac{1}{2}I\,,$$

with the spectral map margin $= 2 - 1/\lambda_{\min}(R)$ [theorem-grade; absolute deviation $\leq 9.9\cdot10^{-10}$ on 85 windows]. *The L-ensemble transformation $Q = \mathsf{E}_N(I+\mathsf{E}_N)^{-1}$ is exact* [rational arithmetic on a 5-atom model; double precision $\leq 7.6\cdot10^{-12}$ on 85 windows]*, and combined with Borodin's complementation $\Pi_N^u = D\,(I - \Pi_{N-1}^{u^\vee})\,D$ yields $\mathsf{E}_N = D(R^{-1} - I)D$ on Y.*

Live census: $\lambda_{\min}(R) - \frac{1}{2}$ ranges from $+4.2 \cdot 10^{-5}$ (running instance: $\lambda_{\min}(R) = 0.500041882$) down to $+5.9 \cdot 10^{-11}$ on the deepest tranche — all positive, no counterexample. Comparison inputs with destroyed source structure violate $R \succ \frac{1}{2}I$ structurally ($\lambda_{\min}(R) - \frac{1}{2} \in [-0.500, -0.083]$). Honest limit: at the deepest instances the double-precision resolution of the dual route ($\sim 5 \cdot 10^{-10}$) exceeds the margins, so the hole coordinate is exact algebra, not a sharper certification instrument.

The pair identities $c = \varepsilon_1\varepsilon_2(R^{-1})_{12}$, $p = 2 - (R^{-1})_{11}$, $q = 2 - (R^{-1})_{22}$ identify the two-point Uvarov block of the companion note with the $(1, 2)$ principal minor of $2I - R^{-1}$ [theorem-grade]. The absolutely continuous hard-edge class is excluded at census grade: the rescaled pair positions $a_j = 2N^2(1 - y_j)$ are family-constant against $\pi^2 j^2/4$ on all 85 windows (maximum deviation $0.33/N$), while the kernel correlation ρ_K decays with fitted exponent 1.4222, contrary to Lubinsky's Bessel limit at fixed microscopic arguments [3], [4, §frozen-placement].

Proposition 2 (Augmented dictionary). *Let* $Z = \left(\begin{smallmatrix} R^{-1} & \widetilde{v} \\ \widetilde{v}^T & 1+\gamma \end{smallmatrix}\right)$, *so* $R^\dagger = Z^{-1}$, *and write* $M^\dagger = R^\dagger - \frac{1}{2}I$. *The following are finite-matrix identities* [theorem-grade; rational arithmetic on toys, graded double precision on 75 windows of the prime-generated family plus two character-twisted families of 42 windows each].

(A1) *The augmented node Gram* $G^\dagger$ *is the sign-conjugate of* $(R^\dagger)^{-1} - I$*: the particle–hole complementation extends by one row.*

(A2) $\mathrm{margin}^\dagger = 2 - 1/\lambda_{\min}(R^\dagger)$ *(deviation* $\le 1.1 \cdot 10^{-11}$ *shallow,* $\le 5.5 \cdot 10^{-10}$ *at the deepest computed row).*

(A3) $L^\dagger \iff R^\dagger \succ \frac{1}{2}I$*, two-sided. Both truth directions occur on real windows: six character-twisted rungs with terminal ratio* $q_N > 1$ *have* $\lambda_{\min}(R^\dagger) - \frac{1}{2} < 0$ *(e.g.* $-9.8 \cdot 10^{-5}$*), exactly as the identity demands.*

(A4) *Sherman–Morrison on the* Y*-block:* $R^\dagger_{YY} = R + (R\widetilde{v})(R\widetilde{v})^T/(1 + \gamma - \widetilde{v}^T R\widetilde{v})$*, and* $R^\dagger_{bb} = 1/(1 + \gamma - \widetilde{v}^T R\widetilde{v})$ *(deviation* $\le 1.2 \cdot 10^{-14}$ *on every computed row, including the deepest). The border is a rank-one insertion at dual-resolvent level.*

(A5) $R^\dagger \succ \frac{1}{2}I$ *if and only if* $R \succ \frac{1}{2}I$ *and* $q^\dagger < 1$*, where* $q^\dagger = \gamma + v^T(I - \mathsf{E}_N)^{-1}v$ *is frame-invariant (equals the squared dual border norm* $u^T H^{-1} u/B$*) and coincides with the standing terminal coordinate. The dictionary* $1 - q^\dagger = (5/7)(1 - q_N)/B$ *holds to relative* $6.7 \cdot 10^{-6}$.

(A6) *The Schur complement of* $M^\dagger$ *onto the border coordinate equals* $(1 - q^\dagger)/\big(2(1 + q^\dagger)\big)$ *globally, and equals the same value inside the local* 3×3 *block of Proposition 3 (quotient* $\le 1.0 \cdot 10^{-11}$*).*

(A7) *Cauchy interlacing:* $\lambda_k(R^\dagger) \le \lambda_k(R) \le \lambda_{k+1}(R^\dagger)$*. The border can only lower the bottom of the dual spectrum (*$\mathrm{margin}^\dagger \le \mathrm{margin}$*). On the running instance,* $0.500041459 \le 0.500041882$.

A formal reconstruction, independent of the dual coordinate, is already machine-checked in Lean: $L^\dagger \iff L^* \wedge \mathrm{Terminal}$ and $L^* \wedge \mathrm{Terminal} \Rightarrow$ positivity of every bordered window matrix through the cap (`RH/Augmented.lean`, `RH/Closure.lean`; axiom audit in `RH/Audit.lean`, consuming only the two named canonical holes). The dual-resolvent dictionary itself is sorry-free since round 373: $L^\dagger \iff R^\dagger \succ \frac{1}{2}I$ under the μ-ONB whitening (`RH/DualResolvent.lean`) [lean-verified]. Haynsworth additivity on the two-rank and mixed J-forms is likewise sorry-free (`RH/Haynsworth.lean`; Section 8). What is *not* formalized, and not proved analytically, is the pair of census premises (P1)–(P2) of Problem 1 for the unbounded family — and, a fortiori, $R^\dagger \succ \frac{1}{2}I$ itself.

4 The critical three-by-three block

Partition $Y = \{y_1, y_2\} \cup C$, with $\{y_1, y_2\}$ the binding pair (folds 2, 4) and C the rest of Y. Write $M = R - \frac{1}{2}I$ and S_N for the 2×2 Schur complement of M onto the pair. Then [exact]

$$M \succ 0 \iff M_{CC} \succ 0 \text{ and } S_N \succ 0\,,$$

and S_N is a ratio of Sylvester-bordered minors of $\det M$ (the rest block sits inside the border; a pair-only truncation is false by 4.07 nats on the running instance and exactly on the rational toy). The off-diagonal of the dual resolvent $A^{-1} = (I - 2R)^{-1}$ is the Casoratian of the two consecutive dual orthonormal polynomials, resolvent-dressed (discrete IIKS identity) [theorem-grade]. The diagonal entries $(A^{-1})_{kk}$ are *not* Casoratian data: their cross share $(A^{-1})_{12}^2/\big((A^{-1})_{11}(A^{-1})_{22}\big)$ has median 0.702 on 85 windows (range 0.46–0.84). No standalone explicit prefactor turns the Casoratian into $\det S_N$ at the sealed bars.

Proposition 3 (Binding). *On the* 74 *resolvable windows of the family* ($\lambda_{\min}(R) - \frac{1}{2} > 10^{-9}$),

$$\text{bind} = \frac{\lambda_{\min}(S_N)}{\lambda_{\min}(R) - \frac{1}{2}} \in [1.0003,\ 1.0605]\,, \quad \textit{median } 1.0058$$

[census]. *After the rank-one border, the* 3×3 *Schur block* $S_3^\dagger$ *of* $M^\dagger$ *on (pair, border) satisfies*

$$\text{bind}^\dagger = \frac{\lambda_{\min}(S_3^\dagger)}{\lambda_{\min}(R^\dagger) - \frac{1}{2}} \in [1.0003,\ 1.0608]\,, \quad \textit{median } 1.0056$$

[census]. *The in-block border Schur equals the global border Schur of* (A6). *The rank-one deformation does not move the pair structure: on the running instance the cross share of* $R^\dagger$ *is* 0.6990 *against* 0.6973 *for* R*; ladder medians* 0.708 *against* 0.702.

On the running instance the three eigenvalues of $S_3^\dagger$ are $(4.16 \cdot 10^{-5},\ 4.90 \cdot 10^{-4},\ 1.76 \cdot 10^{-2})$: the bottom mode is the L^* pair channel, the top mode is the terminal border channel. The critical eigenvector's border mass is $2.5 \cdot 10^{-5}$ there, and $5.8 \cdot 10^{-4}$ at the near-terminal window where $\text{margin}^\dagger / \text{margin} = 0.8716$. The giant operator behaves, at the critical point, like this 3×3 block.

5 The pair straddles the saturation edge

Two independent finite objects describe the dual ensemble's occupation of the cosine grid.

Exact occupation. Let $o_\eta(x_j)$ (resp. $o_{\text{dual}}(x_j)$) be the diagonal of the rank-N η-projection (resp. the rank-$(N-1)$ dual-hole projection). Borodin complementation on the diagonal gives

$$o_\eta(x_j) + o_{\text{dual}}(x_j) = 1 \tag{3}$$

at every union node, with trace ward $\sum_j o_{\text{dual}}(x_j) = N - 1$ [exact; rational arithmetic on the toy, $\le 2.2 \cdot 10^{-11}$ live]. This is the finite counterpart of an equilibrium density versus its upper constraint, with no solver and no asymptotics. On the prime-generated family, fold 1 (the rightmost node) is the unique occupation anomaly: $o_{\text{dual}}(x_1)$ lies 0.25–0.33 below the bulk median on all 85 windows, while the pair folds 2 and 4 are flat (deviation ≤ 0.10).

Obstacle problem. Minimize the discrete energy

$$E(m) = \sum_{j\neq k} m_j m_k \log \frac{1}{|x_j - x_k|} + \sum_j m_j\bigl(-\log u_j^{\vee}\bigr)$$

over $0 \leq m_j \leq 1$, $\sum m_j = N - 1$. The box constraint *is* the Baik–Kriecherbauer–McLaughlin–Miller upper constraint [2]; dropping it piles mass ≥ 92 times the cap onto single nodes. A deterministic projected first-order solver (FISTA plus exact box-simplex water-filling, sealed iteration counts, KKT partition gates) produces, on four sealed instances including the running instance, a dual-void block equal to folds $\{1, 2, 3\}$ [census]. By (3) this is the primal saturated block at the right hard edge.

Proposition 4 (Straddle, census)**.** *The binding pair* $(2, 4)$ *straddles the saturation-block edge* $3 \mid 4$*: fold* 2 *lies inside the saturated zone, fold* 4 *outside. Auxiliary character-twisted families on the same grid have a* different *partition — they occupy fold* 1*, with no fold-*1 *occupation anomaly on* $0/42 + 0/42$ *windows — and a sign-scrambled comparison input is disordered at the edge while failing* $R_{CC} \succ \frac{1}{2}I$ *(*$\lambda_{\min}(R_{CC}) - \frac{1}{2} = -0.4962$*) with the pair block itself still positive. The identity chain is world-blind algebra; the partition, and the positivity, are not.*

The asymptotic equilibrium-measure theorem (BKMM band and saturation limit shape for the field $-\log u^{\vee}$) is not proved [hypothesis, open]. The finite partition is a measurement about this family.

6 The observed exponent

On the 57 calibration windows a Theil–Sen fit of $\log \lambda_{\min}(S_N)$ has slope -3.334, hitting the standing margin exponent 3.332 at distance 0.002 and rejecting an alternative reading 1.65 at distance 1.684 [census]. The diagonal dual-resolvent entries carry the same exponent ($|(A^{-1})_{11}|$ slope $+3.365$, $|(A^{-1})_{22}|$ $+3.292$, $|(A^{-1})_{12}|$ $+3.316$). Weight-side Christoffel exponents (1.79 and 0.38 families) miss by a wide margin: they do not carry the resolvent diagonals.

A dimensionless combination $t_{\mathrm{geo}} = (\lambda_{\min}(R) - \frac{1}{2})\sqrt{(A^{-1})_{11}(A^{-1})_{22}}$ is margin-locked: slope -0.009 on the 57, range $[0.222, 0.287]$ on all 74 resolvable windows [census]. Exact algebra identifies t_{geo} with a function of $\det(S_N/\varepsilon)$ and the cross share; family-constancy is a local-parametrix-class census, not a Gamma-function derivation. The Gamma parametrix of [2, Thm 2.7, Rem 2.8] adjacent to a saturated band remains the natural analytic candidate [hypothesis, open].

An adjugate identity $\det S_N \cdot 4\bigl[(A^{-1})_{11}(A^{-1})_{22} - (A^{-1})_{12}^2\bigr] = 1$ [exact] implies that the wander of $\log\det S_N$ *is* the wander of $-\log\bigl((A^{-1})_{11}(A^{-1})_{22}\bigr)$ (fine-structure correlation $+0.9835$), with the cross-share term five times subleading. The leading common wander lives in the diagonal resolvent data and cancels algebraically in the minor.

7 Relative error-operator form

Two constant-matrix targets fail the census and force a relative statement.

Rest block. Cauchy interlacing on the principal submatrix R_{CC}, and the Schur-complement eigenvalue inequality, give

$$\lambda_{\min}(R_{CC}) - \tfrac{1}{2} \;\geq\; c\left(\lambda_{\min}(R) - \tfrac{1}{2}\right) \qquad \text{with } c = 1 \tag{4}$$

as a finite theorem, gated in both directions on the rational toy and on $74 + 84$ resolvable windows [theorem-grade]. On the prime-generated family the measured ratio lies in $[8.8, 54.2]$

(median 20.3). On the character-twisted families the ratio reaches the interlacing line: minima 1.0 on 11 of 84 windows. Any analytic rest-block clause is a theorem only at $c = 1$; a comfortable factor 8.8 does not transfer.

Leading coefficient. The schematic $S_N = N^{-\alpha}(M_0 + O(N^{-\varepsilon}))$ with M_0 a constant positive matrix is too rigid. The condition number $\kappa_S = \lambda_{\max}(S_N)/\lambda_{\min}(S_N)$ ranges in [6.3, 30.5] (median 13.9), a range ratio $4.88 > 1.5$: the margin direction collapses (bind median 1.0058) but the second eigenvalue wanders by an $O(1)$ factor [census]. The honest form is projective: one controls $S_N/\lambda_{\min}(S_N)$ in a bounded set of positive-definite matrices, or equivalently the error operator $E = P_{\text{true}}P_{\text{model}}^{-1} - I$ of a local Riemann–Hilbert parametrix in a *relative* operator norm on the critical block, not as an additive constant remainder.

No bound on E is proved [open]. Equation (4) at $c = 1$ is the only theorem-grade error control presently available, and it is finite interlacing, not an asymptotic remainder. After rounds 367–373 this relative form is the *stronger* target: the primary question no longer needs a remainder that tracks $N^{-\alpha}$ (Section 8).

8 The minimal inertia form (rounds 367–373)

The schematic $S_N = N^{-\alpha}(M_0 + O(N^{-\varepsilon}))$ of Section 7 is a margin asymptotic. The two-rank inertia cut replaces it by two coarser statements on the rest block $A_0 = R_{N-3} - \frac{1}{2}I$ and the 2×2 Schur matrix $K_2 = I + U^T A_0^{-1} U$, with U the last two dual Christoffel–Darboux columns.

Analytic discount. Write $\operatorname{ind}_-(A)$ for the number of strictly negative eigenvalues of a real Hermitian matrix A. It is enough to prove, uniformly on the admissible family,

$$\operatorname{ind}_-(A_0) \;\le\; 1 \tag{P1}$$
$$\det K_2 \;\le\; -c_0 < 0 \quad \text{on the branch } \operatorname{ind}_-(A_0) = 1. \tag{P2}$$

(P1) is an integer mode count. (P2) is the sign of an $O(1)$ 2×2 source. An asymptotic argument need only beat a *fixed* signature reserve, not the shrinking $N^{-3.33}$ spectral margin of Section 6.

Census on the 74 resolvable windows ($\lambda_{\min}(R) - \frac{1}{2} > 10^{-9}$) [census, round 367]: overload $\operatorname{ind}_-(A_0) \ge 2$ is 0/74; the dichotomy is 45 exact-one (nneg-1) against 29 already positive-definite at rank $N - 3$ (P1 vacuous). On the nneg-1 branch, $-\det K_2 \in [1.157, 1126.389]$, census floor 0.50 holds 45/45, Theil–Sen slope of $\log(-\det K_2)$ against $\log N_w$ is $+0.543$ (flat against the margin rate -3.332), and the restatement correlation against $\log$ margin on the 57 calibration windows is -0.0064. The implication $(\text{P1}) \wedge (\text{P2}) \Rightarrow R \succ \frac{1}{2}I$ holds on all 74 resolvable rows; it is sufficient, not necessary (the vacuous-29 already have $A_0 \succ 0$).

Theorem 1 (Two-rank Haynsworth)**.** *Let A_0 be real Hermitian and invertible with* $\operatorname{ind}_-(A_0) = 1$*, and let $K_2 = I_2 + U^T A_0^{-1} U$. If* $\det K_2 < 0$*, then* $A_0 + UU^T \succ 0$ [lean-verified, `RH/Haynsworth.lean`, sorry-free]*.*

Finite real algebra only; the census premises (P1)–(P2) on windows are *not* asserted. The Lean statement is `haynsworth_two_rank`. On the hole kernel, $A_0 + UU^T = R - \frac{1}{2}I$, so Theorem 1 converts (P1)–(P2) into $R \succ \frac{1}{2}I$. The $\mathrm{R}^\dagger$ dictionary of Proposition 2(A3),(A5) then lifts to $L^\dagger$ once the terminal Schur $q^\dagger < 1$ is available; both directions of $L^\dagger \iff R^\dagger \succ \frac{1}{2}I$ are sorry-free in `RH/DualResolvent.lean`.

Mixed 3×3 form. The border is not a third unaugmented CD column. At dual-resolvent level the Sherman–Morrison insertion of Proposition 2(A4) does supply a third column with an explicit signature. Write $s = R\tilde{v}$, $\mathrm{den} = 1 + \gamma - \tilde{v}^T R\tilde{v}$, and

$$U_{\mathrm{mix}} = \begin{pmatrix} u_{N-3} & u_{N-2} & s \\ 0 & 0 & -1 \end{pmatrix}, \qquad J = \mathrm{diag}\big(1, 1, 1/\mathrm{den}\big).$$

Then [theorem-grade, round 369; Fractions residual exactly 0 with den = 23/30; live residual $\leq 1.04 \cdot 10^{-14}$]

$$M^\dagger \;=\; \mathrm{blkdiag}(A_0, -\tfrac{1}{2}) \;+\; U_{\mathrm{mix}}\, J\, U_{\mathrm{mix}}^T. \tag{5}$$

The generalized Haynsworth identity for invertible Hermitian J,

$$\mathrm{In}\begin{pmatrix} A & U \\ U^T & -J^{-1} \end{pmatrix} = \mathrm{In}(A) + \mathrm{In}(-\Phi) = \mathrm{In}(-J^{-1}) + \mathrm{In}(A + UJU^T),$$

with $\Phi = J^{-1} + U^T A^{-1} U$, is sorry-free (`haynsworth_mixed`) [lean-verified]. The two-rank theorem is the case $J = I_2$.

Census building blocks. Three further finite objects are available to a specialist proof of (P1)–(P2). All three are typed CENSUS or THEOREM-GRADE as marked; none is an extrapolation.

- Wedge-sign stability [census, round 371]: the canonical CD two-form satisfies $\langle \omega, \wedge^2 A_0^{-1} \omega \rangle < 0$ on 45/45 of the nneg-1 branch (range $[-637.7, -1.32]$, zero flips). The second compound matrix therefore has source structure. The capture identity that would turn this into (P2) is algebraically identical to (P2) via the exact expansion of $\det K_2$; it is not independent arithmetic. The $\wedge^2$-subspace share is *not* stable (majority ≥ 0.50 only on 27/45).

- Φ-monotonicity [theorem-grade, round 370]: the matrix-valued Weyl function $\Phi_N(z) = J^{-1} + U^T (A - zI)^{-1} U$ is a block resolvent (Fractions determinant lemma exact at $z = -2$), and $\Phi' \succeq 0$ (live minimum eigenvalue $8.949 \cdot 10^{-9}$ on 18/18 admissible grid nodes). The phase census is two-sided: MAIN 74/74 resolvable rows are classified positive-definite with exact signature balance; the six terminal-dead character-twisted rungs ($q_N > 1$) are classified negative (χ_3 5/5, χ_4 1/1).

- Prüfer dictionary [theorem-grade, round 372]: on the terminal surface, the $\lfloor \theta/\pi \rfloor$-runs of the chain Prüfer phase coincide with the sign-runs of v_2 on 181/181 windows, with the XOR identity $\mathrm{sign}(c_t) = \mathrm{sign}(w \cdot x \cdot v_2)$ on the same 181. The dictionary is a theorem; the one-defect extraction that would have closed the terminal half from a single turning is [refuted].

Documented dead ends. Four internal routes are closed, each at a named place.

- No 3×3 Jacobi Casoratian [refuted, round 370]: dual p and dual-measure Cauchy q satisfy the three-term recurrence; the border Cauchy sequence r_n is not a homogeneous dual solution (residual $3.615 \cdot 10^{-2}$ on the running instance). There is no (p, q, r) Casoratian, so a matrix-Weyl index that would replace Haynsworth by an oriented winding does not linearize.

- Phase restatement [refuted, round 370]: the real-line census index $\Xi = \mathrm{nneg}\,\Phi(0)$ *is* Haynsworth inertia in other coordinates. The phase language does not replace the inertia cut.

- $O(1)$ buffer of mass counting [refuted, round 366]: proportional Markov–Stieltjes counting in the pair gap predicts occupation $Z = 0$ ($nM_I/U = 2.40 \cdot 10^{-5}$ on the running instance); measured $Z = 1$ lives in the $O(1)$ sandwich buffer, which cannot separate 0 from 1 (0/74 scaled-band candidates).

- Edge pinning [refuted, round 363]: dual OP zeros are not exponentially node-pinned in the saturated block $\{1, 2, 3\}$ (Hurwitz maximum 0.462 against bar 0.15). The zone is dual-void / primal-saturated; a BKMM pinning argument at the right hard edge does not apply.

Lean landing pad. Sorry census: five (two arithmetic holes `lstar_canonical`, `terminal_q_canonical`, now the alternative route; the named dictionary `pair_terminal_dictionary`; the opacity bridge `mainWindow_iff_builtFromPrimeSource`; the remaining Level-C arch statement `arch_elementwise_stabiliza`
The entire R† layer — Haynsworth, the dual-resolvent dictionary (A2)–(A5), (A7-min), the window bridge, the one-defect absorption, and the r412 graph-resolvent identities — consumes no `sorryAx`. A proof of (P1) and (P2) for all selected windows beyond the certified head lands, via the kernel-checked chain, at window-local master positivity. That is a landing pad, not a theorem of the family. The live mincut is the named Prop `selected_augDualResolvent_gt_half`.

9 Problem statement

Problem 1 (Minimal inertia of the rest block)**.** *Let (μ, ν) run over the admissible family of [4], with dual kernel R as in Section 2, rest block $A_0 = R_{N-3} - \frac{1}{2}I$, and $K_2 = I + U^T A_0^{-1} U$ the two-rank Schur matrix of Section 8. Prove* (P1) *and* (P2) *uniformly for the unbounded family: at most one negative inertia direction in A_0, and $\det K_2 \le -c_0 < 0$ on the nneg-1 branch. Which theorem in the Riemann–Hilbert analysis of discrete orthogonal polynomials — in particular, which statement in the saturation regime of Baik–Kriecherbauer–McLaughlin–Miller* [2, Thm 2.7, Remark 2.8] *— supplies these two coarser facts? The target is an integer mode count and the sign of an $O(1)$ minor, not a rate.*

By Theorem 1 a proof of Problem 1 gives $R \succ \frac{1}{2}I$. The R† layer then lifts to $L^\dagger$ once the terminal Schur is in hand (Proposition 2(A3),(A5), Lean-verified). The mixed identity (5) is the bordered analogue: the same Haynsworth additivity, with $J = \mathrm{diag}(1, 1, 1/\mathrm{den})$, applies to $M^\dagger$.

Problem 2 (Stronger saturation form)**.** *In the notation of Definition 1, which local parametrix at a hard edge adjacent to a saturated band guarantees*

$$R^\dagger \;\succ\; \tfrac{1}{2}I \tag{6}$$

uniformly, equivalently that the critical 3×3 Schur block of $R^\dagger - \frac{1}{2}I$ (binding pair plus border) stays positive-definite? This is the pre-367 target: a margin asymptotic $S_N = N^{-\alpha}(M_0 + \cdots)$ with relative error control. It implies Problem 1 and is strictly stronger.

The finite input to either theorem is assembled above: reciprocal weight (1); occupation duality (3); a census-grade saturated block $\{1, 2, 3\}$ with the binding pair astride the edge $3 \mid 4$; the dichotomy $\mathrm{ind}_-(A_0) \le 1$ on 74/74 with (P2) on the 45-row nneg-1 branch; mixed form (5); observed rate $N^{-3.332}$ (now optional); local-parametrix datum t_{geo} margin-locked in $[0.222, 0.287]$; error control in relative/projective form, with rest-block theorem only at $c = 1$. The AC/Bessel class is the wrong class [3]. A Gamma-class derivation at saturation adjacency remains a candidate for the *stronger* form and is open. The cheaper target is (P1)–(P2).

10 The graph-resolvent form (rounds 407–412)

Rounds 407, 409 and 411 produced three equivalent finite-algebra dictionaries for the rest-block count (P1). Round 412 records them as sorry-free matrix identities in `rh/lean/RH/GraphResolvent.lean`. Nothing below asserts (P1) or $R^\dagger \succ \frac{1}{2}I$ on the family; the mincut remains the named Prop `selected_augDualResolvent_gt_half`. This is a coordinate consolidation, not a claim.

Dictionary. Let $C \succ 0$ be a real positive-definite Gram (on the campaign, the μ-only dual CD Gram on the hole set Y). The *graph resolvent* is

$$R \;=\; C(I+C)^{-1} \;=\; (I+C^{-1})^{-1} \;=\; I-(I+C)^{-1}. \tag{7}$$

The first form is the r407 intertwiner; the second is the r356 dual resolvent of the L-ensemble kernel $E = C^{-1}$ (r409 graph identity). Well-definedness, $0 \prec R \prec I$, and the spectral law $\mathrm{eig}(R) = \lambda/(1+\lambda)$ are finite algebra [lean-verified]. The half-shift identity

$$R - \tfrac{1}{2}I \;=\; \tfrac{1}{2}\,(I+C)^{-1}(C-I) \tag{8}$$

is an equality of matrices. Because $(I+C)^{-1} \succ 0$, Sylvester gives

$$\mathrm{ind}_-(R-\tfrac{1}{2}I) \;=\; \mathrm{ind}_-(C-I). \tag{9}$$

The factorization $I - C^{-1} = C^{-1}(C-I)$ is *not* a congruence. The true sandwich is $C^{-1}(C-I)C^{-1}$, equivalently the simultaneous eigenbasis, and yields the Möbius law

$$\mathrm{ind}_-(C-I) \;=\; \mathrm{ind}_-(I-C^{-1}). \tag{10}$$

If a rectangular factor satisfies $\mathfrak{T}^*\mathfrak{T} = C^{-1}$ (r409: the min-norm Birkhoff extension $\mathfrak{T}_0$), the energy split is

$$\|\mathfrak{T}\| \;=\; 1/\sqrt{\lambda_{\min}(C)}, \qquad \|\mathfrak{T}\| \le 1 \iff C \succeq I, \tag{11}$$

and "at most one singular value > 1" is $\mathrm{ind}_-(I - \mathfrak{T}^*\mathfrak{T}) \le 1$, which is (9)–(10) [lean-verified].

Four coordinates for (P1). The integer count (P1) is the same statement in four languages:

$$\begin{aligned} \mathrm{ind}_-(R-\tfrac{1}{2}I) \;&\le\; 1, && \text{(inertia)}\\ \lambda_2(C) \;&\ge\; 1, && \text{(sampling)}\\ \mathfrak{T}_0 \text{ has at most one singular value} &> 1, && \text{(contraction)}\\ C \succeq I \text{ except in at most one direction.}& && \text{(energy split)} \end{aligned}$$

The zero-defect face $C \succ I$ together with the A5 Schur $q^\dagger < 1$ lifts to $R^\dagger \succ \frac{1}{2}I$ [lean-verified, `augDualResolvent_gt_half_of_C_gt_one`]. The one-defect lift at the same depth is blocked by A7-min (a one-defect of R cannot be absorbed into $R^\dagger$ at the same size). Identifying the campaign Gram C with E^{-1} on a μ-ONB transcription is the named Prop `GraphResolventIsLEnsembleInv`, the same class as `P1EqCapInertia`; it is not a `sorry`.

Running-instance margins. On the frozen window $z = 16$ ($S = 367$, $N = 184$, $|Y| = 104$) [census, rounds 407–411]: $C_{\min} = 0.85712$, $\lambda_2(C) = 1.00018$, $r_{\min}(R) = 0.46153 = C_{\min}/(1+C_{\min})$, $\|\mathfrak{T}_0\| = 1.08014 = 1/\sqrt{C_{\min}}$. Exact identity saturation $\|\mathfrak{T}_0\| = 1$ on already-PD windows is [refuted, round 411]: $C_{\min} - 1 = +4.57 \cdot 10^{-8}$ at $k_z = 42$, $+4.9 \cdot 10^{-9}$ at $k_z = 130$, core-42 PD in $[+1.33 \cdot 10^{-7}, +7.54 \cdot 10^{-5}]$. These numbers certify the dictionary; they do not prove (P1) off the certified head.

Closed internal languages. Table 1 lists seventeen named routes to (P1) closed between DCCXLVI and DCCLXXVIII. Each line is a sealed probe record, not an extrapolation. Two further r409 closures (literal $\Theta = C^T\Phi_N C$; unfitted Krein) sit in the same bin and are omitted from the count only because they attack the graph factor, not (P1) itself.

Lean layer. `RH/GraphResolvent.lean` is sorry-free. The axiom audit (`RH/Audit.lean`, section (m)) prints `[propext, Classical.choice, Quot.sound]` on the identities (7)–(11), on the four-coordinate bridge, and on the zero-defect composition with $R^\dagger$. The pilot `sorry` census stays five. The live remainder is still the CD identification plus the depth gap $N-3$ versus the cap, i.e. the named mincut `selected_augDualResolvent_gt_half`.

Table 1: Seventeen closed languages for (P1), DCCXLVI–DCCLXXVIII. Verdicts are probe-local. No family theorem is claimed.

Language	Round	Verdict
$L^\infty/d_{\mathrm{arm}}$ majorant	r381, DCCXLVI	too coarse
Fold-parity pairing as SATZ	r383, DCCXLIX	residual 0.512
Finite-head Chebyshev / family-uniform Z	r383/r386, DCCL–DCCLI	χ mutants
Gershgorin/Schur assist	r387, DCCLII	$k^* \sim 0.11\,N$
OSC-Geronimus / Neumann R_2	r388, DCCLIII	$\lvert\mathrm{quad}\rvert/\lvert\mathrm{lin}\rvert \sim 0.86$
Checkerboard / M-matrix	r394, DCCLIX	agree 0.504
Three-gap mask	r395, DCCLX	$n_{\mathrm{uniq}} \in [8, 18]$
Isolation of the fold mask	r396, DCCLXI	pair count does not $\to 0$
Source Weyl energy decay	r399, DCCLXV	E^{bulk} grows
Bulk FORM_T / FORM_P	r400, DCCLXVII	rank-1 / phase predictor
(P1) as construction class	r403, DCCLXVIII	REFUTED
Moments / Euler source Gram	r404, DCCLXXI	residual 209
Ones-mode / lift overflow	r404–r405, DCCLXXI–II	$\cos 0.007$; dead χ keep $\lVert c\rVert < 1$
ν-rank one / Loewner sandwich	r407, DCCLXXIII	$N_\nu = 104$
Nyquist density $\frac{1}{2}$ / $\operatorname{diag} C \approx 1$	r408, DCCLXXV	density 0.283
Testpoly / Fourier / Bernstein / SEQ	r410, DCCLXXVII	factor 90; off-fraction 0.968
Identity saturation $\lVert\mathfrak{T}_0\rVert = 1$	r411, DCCLXXVIII	$C_{\min} - 1 = +4.57 \cdot 10^{-8}$

11 The edge balance chain (rounds 417–425)

Rounds 417–424 produced eight finite identities for the edge Schur scalar sch < 0; round 425 reduced the last chain step $\gamma < 1$ to the already-named terminal ratio $q_N < 1$. Round 426 records the abstract halves in `rh/lean/RH/EdgeBalance.lean`. Nothing below asserts sch < 0 cofinally, nor $R^\dagger \succ \frac{1}{2}I$, on the unbounded family. This is a coordinate consolidation, not a claim.

Eight identities. Write $A_0 = R_{N-3} - \frac{1}{2}I$ on the hole nodes, U_{CD} the last two dual Christoffel–Darboux columns, s the Sherman–Morrison border, and den > 0 the border scale. The mixed 3×3 is

$$\Phi = \begin{pmatrix} K_2 & c_0 \\ c_0^T & \varphi_{bb} \end{pmatrix}, \quad K_2 = I + U_{\mathrm{CD}}^T A_0^{-1} U_{\mathrm{CD}}, \quad \varphi_{bb} = \mathrm{den} - 2 + s^T A_0^{-1} s,$$

and sch is the Schur complement of K_2 in Φ. After the 2×2 Sylvester map of K_2, write $(\tilde a, \tilde b)^T = D_2^T c_0$ for the *unnormalized* couplings and $\tau^2 = \tilde a^2 + \tilde b^2$.

1. **Woodbury and the unnormalized chart** [exact, lean-verified].

$$\mathrm{sch} = \mathrm{den} - 2 + s^T (A_0 + U_{\mathrm{CD}} U_{\mathrm{CD}}^T)^{-1} s = \varphi_{bb} - c_0^T K_2^{-1} c_0.$$

The first form is the r406 discriminant Δ at $J = I$: sch $= (\mathrm{den} - 1) - \Delta$. At den $= 1$ one has sch < 0 if and only if the one-defect update is positive definite. The chart of $J_2 = \operatorname{diag}(\varepsilon_a, \varepsilon_b)$ with $\varepsilon \in \{\pm 1\}$ is the 3×3 Schur sch $= \varphi_{bb} - (\varepsilon_a \tilde a^2 + \varepsilon_b \tilde b^2)$, because $J_2^{-1} = J_2$:

$$\begin{aligned} \text{P1 } (\varepsilon = (-1, +1))&\colon \mathrm{sch} = \varphi_{bb} + \tilde a^2 - \tilde b^2, \\ \text{vacuous } (\varepsilon = (+1, +1))&\colon \mathrm{sch} = \varphi_{bb} - (\tilde a^2 + \tilde b^2), \\ \text{tot } (\varepsilon = (-1, -1))&\colon \mathrm{sch} = \varphi_{bb} + \tilde a^2 + \tilde b^2. \end{aligned}$$

Rational toys: sch $= -2/3$ (P1) and $-7/6$ (vacuous).

2. **One balance** [exact, lean-verified]. On both charts $\varphi_{bb} = c_J + \Sigma$ with $c_J = \mathrm{den} - 2$ and $\Sigma = s^T A_0^{-1} s$. Flagship $z = 16$: $-0.0856 = -0.3989 + 0.3133$.

3. **Vacuous τ^2-separator** [exact, lean-verified]. On the vacuous chart, $\mathrm{sch} < 0 \iff \tau^2 > \varphi_{bb}$. In particular $\varphi_{bb} < 0$ forces $\mathrm{sch} < 0$ for every coupling. On the $\varphi_{bb} > 0$ class this is the exact life/death cut (live overflow iff $\tau^2 > \varphi_{bb}$; dead vacuous χ_3-23/33, χ_4-20 iff $\tau^2 < \varphi_{bb}$).

4. **Branch τ-dynamics** [census]. Vacuous $|\tau|$ slope -1.24 (vanishes); P1 slope $+0.05$ (flat). The r417 decay was vacuous-driven. Six core overflow windows live now by $\tau^2 > \varphi_{bb}$ (margins 0.0015–0.046) and do not cross $\varphi_{bb} = 0$ (slope -0.003); they are finite exceptions.

5. **Stieltjes form of Σ** [exact]. Via the r407 dictionary $A_0^{-1} = 2(C+I)(C-I)^{-1}$,

$$\Sigma = 2\|s\|^2 + 4\, s^T(C-I)^{-1}s = 2\sum_i \mathrm{occ}_i\, g(\lambda_i),$$

the Stieltjes transform of the occupation measure at $z = 1$. Near-1 mass stays $O(0.01)$: n_{eff} growth is bulk broadening, not pole-creep. No elementary closed occupancy density (one-pole relative error 0.73; Cauchy two-moment 0.95).

6. **den formula** [exact, lean-verified].

$$\mathrm{den} = 1 + \gamma - v_t \cdot s, \qquad \gamma = \|b\|^2/B_w, \qquad B_w = S_{N-2} + \tfrac{5}{7},$$

and $\mathrm{den} < 2 \iff \gamma < 1 + v_t \cdot s$. If $\gamma < 1$ and $v_t \cdot s \geq 0$ then $\mathrm{den} < 2$ even at vanishing correction. Census: $\mathrm{den} \in [1.460, 1.652]$ on all 42 (gap to 2 at least 0.348); $v_t \cdot s \in [0.061, 0.096]$.

7. **b-Parseval** [exact; identification named in Lean]. $b_k = \langle \sigma, \hat{\pi}_k^{\mu} \rangle$ with σ the signed border measure, so $\|b\|^2$ is the μ-side Parseval budget (w9 residual $4 \cdot 10^{-14}$; $\mathbb{Q}$-toy $b = (3/5, 4/5)$, $\|b\|^2 = 1 < 16/9$). Termwise $w_k = b_k^2/\rho_k$ is not uniform ($\sim 43\%$ of terms > 1).

8. **Floor and three saturation scales** [census]. Selected $a_k = 2^k$ prefers $R = -\varphi_{bb} = R_\infty + c\,k^{-\alpha}$ with $R_\infty \approx +0.030$ against decay-to-0 (ΔAIC 6.9) and against $c/\log k$ (killed, ΔAIC 18.4). EXT-vacuous sits independently in $[0.0285, 0.0373]$. Three scales, not one exponent: budget gap $2 - \mathrm{den} \sim O(0.5)$ flat; razor $C_{\min} - 1 \sim 10^{-7}$ with $\|\mathfrak{T}_0\| - 1 = -(C_{\min} - 1)/2$ exactly (ratio 1.0000); edge reserve R of order 0.03.

Branch map. Exact separators, not a fitted partition:

branch	chart / K_2	live criterion
P1 (28/42 core)	$J_2 = \mathrm{diag}(-1, +1)$	$\mathrm{sch} = \varphi_{bb} + \tilde{a}^2 - \tilde{b}^2 < 0$ (hyperbola; danger axis large $\lvert\tilde{a}\rvert$)
vacuous (14/42)	$J_2 = I_2$	$\varphi_{bb} < 0$, or $\tau^2 > \varphi_{bb}$ on the $\varphi_{bb} > 0$ overflow class
tot / dead χ	$J_2 = -I$ or $\mathrm{sch} > 0$	vacuous death: $\varphi_{bb} > 0$ and $\tau^2 < \varphi_{bb}$; P1 death: τ-terms, $\varphi_{bb} < 0$ retained

Round 425: $\|b\|^2 \leq S$ is the kernel Loewner $K_\mu \preccurlyeq K_{\mu-\nu}$ on $P_{<N}$ [exact]; $\|b\|^2 < B_w$ is that identity plus $q_N < 1$. On the 42-rung $q_N < 1$ census this composes to $\gamma < 1$. Dead χ have $q_N > 1$ and need kernel slack; they keep $\gamma < 1$ nonetheless.

Two open limit questions. The identities above are finite. What remains, for the unbounded selected sequence $a_k = 2^k$, are two existence statements. They are the cheapest remaining targets; neither is a Riemann–Hilbert steepest-descent, and neither is a statement about the Riemann hypothesis.

Problem 3 (cofinal $q_N < 1$)**.** *On the selected sequence, prove $q_N < 1$ (equivalently $S < B_w = S_{N-2} + \frac{5}{7}$) for all large k. Together with the Loewner identity $\|b\|^2 \le S$ this yields $\gamma < 1$, hence* $\mathrm{den} < 2$ *by the formula of item 6, even at $v_t \cdot s = 0$. The 42-rung census is already certified (v960); cofinality is the named mincut `selected_augDualResolvent_gt_half`.*

Problem 4 (constructed Σ_∞)**.** *Construct* $\mathrm{den}_\infty < 2$ *and* Σ_∞ *with*

$$\limsup_k \Sigma_k \;<\; 2 - \mathrm{den}_\infty \;=\; |c_{J,\infty}|,$$

*equivalently a strictly positive floor $R_\infty > 0$ for $R = -\varphi_{bb}$. The occupation Stieltjes form is exact; a closed occupancy density is refuted; near-*1 *mass is stable* $O(0.01)$*. The census prefers* $R_\infty \approx 0.030$ *(two independent lines: selected AIC, EXT-vacuous band). No k_0 is proved. Depth* $k = 10$ *border-fails computationally.*

Lean layer. `RH/EdgeBalance.lean` is sorry-free. The axiom audit (`RH/Audit.lean`, section (n)) prints `[propext, Classical.choice, Quot.sound]` on the Woodbury-sch corollary, the chart trichotomy, the τ^2-separator, the den formula and the γ-bridge $\|b\|^2 \le S < B_w \Rightarrow \gamma < 1 \Rightarrow$ $\mathrm{den} < 2$. Named Props `BorderIsMuParseval`, `BorderLoewnerLeS`, `QNLtOne` are hypotheses, not holes: they name the window transcription of items 7–8 and Problem 3. The pilot `sorry` census stays five.

12 What a proof would give, and what it would not

A proof of Problem 1 for the unbounded family would give, in order: $R \succ \frac{1}{2}I$ by Theorem 1 [lean-verified finite algebra]; then $R^\dagger \succ \frac{1}{2}I$ once $q^\dagger < 1$, equivalently $L^\dagger$, by Proposition 2(A3),(A5) [theorem-grade and lean-verified]; $L^\dagger \Rightarrow L^* \wedge \text{Terminal}$ and $L^* \wedge \text{Terminal} \Rightarrow$ positivity of every bordered window matrix through the cap [Lean, sorry-free on the reconstruction layer, axiom audit `RH/Audit.lean`]. That is window-local master positivity. A proof of the stronger form (6) would give the same chain without a separate terminal input. Neither is an asymptotic law, a bound on the Riemann ξ-function, or a statement about the Riemann hypothesis in either direction.

What is on the table now is a finite census (at most 85 windows for the hole kernel, 75 plus one deep row for the border, four sealed obstacle instances for the partition, 74 resolvable rows for the two-rank dichotomy, 181 windows for the Prüfer dictionary) together with exact finite-matrix identities and a sorry-free R† Lean layer. No Riemann–Hilbert steepest-descent analysis has been carried out. The equilibrium-measure limit, the Gamma parametrix, and the error operator E are open, and they are no longer the cheapest target. Comparison inputs show that the identities survive source destruction while positivity and the saturated partition do not — so an existence proof for (P1)–(P2) must consume the construction, not merely the dual algebra.

Table 2 records the running instance. Every number is a sealed probe record. Numbers not in those records are omitted.

Claim boundary. Research documentation. Not a proof of L^*, of $L^\dagger$, of (P1)–(P2), or of $R^\dagger \succ \frac{1}{2}I$ for the unbounded family. Not evidence for or against the Riemann hypothesis in either direction.

Table 2: Running instance $z = 16$ ($S = 367$, $N = 184$, pair folds $(2, 4)$). Dual quantities: round 356/359. Occupation and obstacle: round 360. Bordered quantities: round 362. Two-rank inertia: round 367.

quantity	value
$\lambda_{\min}(R)$	0.500041882
$\lambda_{\min}(R^\dagger)$	0.500041459
margin / margin†	$1.6752 \cdot 10^{-4}$ / $1.6582 \cdot 10^{-4}$
bind / bind†	1.0029 / 1.0028
cross share / share†	0.6973 / 0.6990
t_{geo}	0.2646
o_{dual} at folds $1, 2, 4$	0.1497, 0.5118, 0.5137
QP dual-void block (KKT gap)	$\{1, 2, 3\}$ (+0.092)
$q^\dagger$ / border Schur	0.933044 / $1.7319 \cdot 10^{-2}$
$\text{ind}_-(A_0)$ / $\det K_2$	1 / -5.0389
$-\det K_2$ (nneg-1 family)	[1.157, 1126.389]

References

[1] A. Borodin, *Duality of orthogonal polynomials on a finite set*, J. Statist. Phys. **109** (2002), 1109–1120.

[2] J. Baik, T. Kriecherbauer, K. T.-R. McLaughlin and P. D. Miller, *Discrete Orthogonal Polynomials: Asymptotics and Applications*, Ann. of Math. Studies **164**, Princeton Univ. Press, 2007. Theorem 2.7 and Remark 2.8.

[3] D. S. Lubinsky, *A new approach to universality at the edge of the spectrum*, Int. Math. Res. Papers (2008).

[4] S. Hamann, *A strict polynomial subordination inequality for two explicit prime-generated discrete measures (open problem)*, TFPT research repository, `rh/problem/lstar_problem.tex`, August 2026.